

VPA IGCSE ICT Year 10 Semester 2 — Course Overview

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Victoria Park Academy | IGCSE ICT (0417) | Year 10 Semester 2

Duration: 18 weeks | Focus: Consolidation of all IGCSE topics + Mock Examination Preparation

This semester bridges the gap between topic completion and examination readiness, ensuring students are confident across all three papers.

Student Profile

Target Cohort: Year 10 students (ages 14–15) preparing for IGCSE ICT examination in Year 11.

Entry Profile: Students have completed Year 10 Semester 1, covering:

Document production (word processing, formatting, mail merge)

Data manipulation (spreadsheets, formulas, charts, pivot tables)

Presentations (slide design, transitions, multimedia)

Data analysis (modelling, what-if analysis, goal seek)

Web authoring (HTML, CSS, multi-page sites)

Databases (design, relationships, queries, forms, reports)

Networks and security (protocols, encryption, threats)

Exit Profile: By the end of this semester, students will:

Demonstrate fluency across all IGCSE ICT practical skills

Complete timed examination-style tasks under mock conditions

Identify personal areas of strength and weakness through gap analysis

Develop effective examination technique and time management

Be prepared to enter Year 11 with a clear revision and improvement plan

学生概况：本课程面向10年级学生（14–15岁），为11年级IGCSE ICT考试做准备。学生已完成10年级第一学期，涵盖文档制作、数据处理、演示文稿、数据分析、网页制作、数据库和网络与安全。本学期结束时，学生将熟练掌握所有IGCSE ICT实践技能，完成模拟考试条件下的计时任务，通过差距分析识别个人优势和弱点，发展有效的考试技巧和时间管理，并为进入11年级制定清晰的复习和改进计划。

18-Week Curriculum Map

Week	Focus	Key Activities	Assessment
1	Document Production Consolidation	Timed document formatting task; styles, mail merge, tables of contents	Diagnostic test (Paper 2 style)
2	Data Manipulation Consolidation	Complex spreadsheet with nested IF, VLOOKUP, conditional formatting	Peer assessment; teacher feedback

Week	Focus	Key Activities	Assessment
3	Presentation Consolidation	Multi-slide presentation with transitions, animations, embedded media	Self-assessment checklist
4	Data Analysis Consolidation	What-if analysis, goal seek, pivot tables, chart creation	Timed practical task
5	Web Authoring Consolidation	Multi-page website with CSS, navigation, forms, embedded media	Peer code review
6	Database Consolidation	Relational database design, SQL queries, forms, reports	Timed practical task
7	Mixed Practical Skills 1	Combined task: document + spreadsheet + presentation	Formative assessment
8	Mixed Practical Skills 2	Combined task: web + database + data analysis	Formative assessment
9	Consolidation Review & Gap Analysis	Self-assessment; personalised revision plan creation	Individual progress report
10	Mock Paper 1 — Theory	Full timed mock examination (2 hours)	Mock Paper 1 (graded)
11	Mock Paper 1 Feedback	Mark scheme review; error analysis; targeted revision	Improvement plan
12	Mock Paper 2 — Document Production	Full timed mock examination (2 hours 30 minutes)	Mock Paper 2 (graded)
13	Mock Paper 2 Feedback	Mark scheme review; common errors; technique refinement	Improvement plan
14	Mock Paper 3 — Web & Database	Full timed mock examination (2 hours)	Mock Paper 3 (graded)
15	Mock Paper 3 Feedback	Mark scheme review; SQL debugging; HTML/CSS refinement	Improvement plan
16	Gap Analysis & Targeted Revision	Personalised revision based on mock performance data	Progress tracking
17	Final Examination Technique	Time management drills; question strategy; stress management	Timed practice
18	Final Preparation & Confidence Building	Light revision; peer teaching; confidence activities	End-of-semester report

Assessment Calendar

Assessment	Week	Format	Weighting
Diagnostic Test	1	Paper 2 style practical (1 hour)	Formative
Consolidation Tasks (×6)	1 – 6	Timed practical tasks	Formative
Mixed Skills Tasks (×2)	7 – 8	Combined practical tasks	Formative

Assessment	Week	Format	Weighting
Self-Assessment & Gap Analysis	9	Portfolio review + reflection	Formative
Mock Paper 1 (Theory)	10	2-hour written examination	Summative (15%)
Mock Paper 2 (Document Production)	12	2.5-hour practical examination	Summative (15%)
Mock Paper 3 (Web & Database)	14	2-hour practical examination	Summative (15%)
Final Examination Technique Test	17	Timed mixed skills task	Formative

EAL Checklist

English as an Additional Language Support:

- ☐ Bilingual glossary provided for all technical terms (EN/中文)
- ☐ Examination instructions read aloud in addition to written form
- ☐ Extra time (25%) available for timed assessments upon application
- ☐ Word banks provided for extended-response questions
- ☐ Visual aids and diagrams encouraged in written responses
- ☐ Peer translation support during consolidation (not during mocks)
- ☐ Simplified English versions of case studies and scenarios
- ☐ One-to-one check-ins with EAL coordinator every 3 weeks

EAL支持清单：为所有技术术语提供双语词汇表（英/中）；口头朗读考试说明；可申请额外25%时间；为扩展回答题提供词汇库；鼓励书面回答中使用视觉辅助和图表；巩固阶段提供同伴翻译支持（模拟考试期间不提供）；提供案例研究和场景的简化英语版本；每3周与EAL协调员进行一对一检查。

Differentiation

Bronze Level — Developing

Support	Description
Scaffolded Tasks	Step-by-step worksheets with partially completed examples for each practical skill
Extended Time	Additional 15 minutes for consolidation tasks; 25% extra for assessments
Peer Support	Pairing with Silver/Gold students for guided practice; structured peer tutoring
Simplified Instructions	Chunked instructions with visual cues; numbered steps with checkboxes
Focus Areas	Prioritise core skills: basic document formatting, simple formulas, basic HTML/CSS, simple queries

Silver Level — Secure

Support	Description
Standard Tasks	Full examination-style tasks with standard time limits
Independent Practice	Self-directed consolidation with teacher check-ins
Extension Opportunities	Optional challenge tasks: macros, advanced CSS, complex SQL with JOINS
Peer Teaching	Opportunities to explain concepts to Bronze students (reinforces learning)
Focus Areas	Develop fluency across all topics; refine examination technique; build speed

Gold Level — Advanced

Support	Description
Stretch Tasks	Advanced challenges: VBA scripting, JavaScript interactivity, database normalisation to 3NF, responsive design
Leadership Roles	Peer tutor for Bronze/Silver students; lead study groups; create revision resources
Competition Preparation	Prepare for Bebras, UKMT, or coding competitions alongside IGCSE preparation
Research Projects	Independent investigation into emerging ICT topics (AI, quantum computing, blockchain)
Focus Areas	Achieve examination excellence; develop transferable skills for IB DP Computer Science

50+ Term Glossary for Semester 2

Term	Definition	Paper
Mail Merge	Automating document creation by combining a template with a data source	P2
Nested IF	An IF function inside another IF function for multiple conditions	P2
VLOOKUP	Vertical lookup function to search for a value in the first column of a table	P2
Conditional Formatting	Automatically changing cell appearance based on cell values	P2
Pivot Table	Dynamic table that summarises and analyses data from a larger dataset	P2
Goal Seek	Tool that finds the input value needed to achieve a desired output	P2
What-If Analysis	Exploring different scenarios by changing input values in a model	P2
Absolute Reference	Cell reference that remains fixed when copied (\$A\$1)	P2
Relative Reference	Cell reference that adjusts when copied (A1)	P2

Term	Definition	Paper
Slide Master	Template that controls the default layout and appearance of slides	P2
Transition	Animation effect between slides in a presentation	P2
Animation	Effect applied to individual elements within a slide	P2
HTML	HyperText Markup Language — the standard language for creating web pages	P3
CSS	Cascading Style Sheets — controls the visual presentation of HTML elements	P3
Semantic HTML	Using HTML tags that convey meaning (header, nav, article, footer)	P3
Responsive Design	Design approach that makes web pages render well on all screen sizes	P3
Form Validation	Ensuring user input meets specified criteria before submission	P3
Primary Key	Unique identifier for each record in a database table	P3
Foreign Key	Field that links one table to another in a relational database	P3
Normalisation	Process of organising data to reduce redundancy and improve integrity	P3
SQL	Structured Query Language — used to manage and query relational databases	P3
SELECT	SQL command to retrieve data from a database	P3
WHERE	SQL clause to filter records based on a condition	P3
JOIN	SQL operation to combine rows from two or more tables	P3
Data Integrity	Accuracy and consistency of data throughout its lifecycle	P3
Validation	Checking that data is reasonable and meets specified rules	P3
Verification	Checking that data has been entered correctly (e.g., double entry)	P3
Phishing	Fraudulent attempt to obtain sensitive information by disguising as a trustworthy entity	P1
Encryption	Converting data into a code to prevent unauthorised access	P1
Firewall	Network security system that monitors and controls incoming/outgoing traffic	P1
Malware	Software designed to disrupt, damage, or gain unauthorised access to systems	P1

Term	Definition	Paper
Bandwidth	Maximum rate of data transfer across a network path	P1
Latency	Delay between a user's action and the system's response	P1
Cloud Computing	Delivery of computing services over the internet	P1
IoT	Internet of Things — network of physical devices embedded with sensors and software	P1
VPN	Virtual Private Network — extends a private network across a public network	P1
Cookie	Small piece of data stored on the user's computer by a web browser	P1
Digital Footprint	Trail of data created by an individual's online activity	P1
Copyright	Legal right protecting creators of original works from unauthorised use	P1
Creative Commons	Licences that allow creators to specify how others can use their work	P1
Algorithm	Step-by-step procedure for solving a problem or completing a task	P1
Pseudocode	Simplified programming notation using natural language and basic programming structures	P1
Flowchart	Diagram representing the sequence of steps in an algorithm	P1
Variable	Named storage location that holds a value which can change	P1
Loop	Programming structure that repeats a sequence of instructions	P1
Conditional	Programming structure that executes different instructions based on a condition	P1
Binary	Number system using only 0 and 1, the language of computers	P1
Pixel	Smallest unit of a digital image, short for "picture element"	P1
Resolution	Number of pixels in an image, typically expressed as width × height	P1
Lossy Compression	Data compression that permanently removes some information to reduce file size	P1
Lossless Compression	Data compression that reduces file size without losing any information	P1
MAC Address	Unique identifier assigned to a network interface controller	P1

Term	Definition	Paper
IP Address	Unique address that identifies a device on the internet or a local network	P1
DNS	Domain Name System — translates human-readable domain names to IP addresses	P1
SSL/TLS	Protocols that encrypt data transmitted between web browsers and servers	P1
Two-Factor Authentication	Security process requiring two different authentication methods	P1
Social Engineering	Psychological manipulation of people into performing actions or divulging information	P1
Backup	Copy of data stored separately to protect against data loss	P1
Disaster Recovery	Plan for restoring systems and data after a catastrophic event	P1
UPS	Uninterruptible Power Supply — provides emergency power during outages	P1
BIOS	Basic Input/Output System — firmware that initialises hardware during booting	P1
Operating System	Software that manages computer hardware and software resources	P1
Open Source	Software with source code that anyone can inspect, modify, and enhance	P1
Proprietary Software	Software owned by an individual or company with restricted usage rights	P1
Agile Development	Iterative approach to software development that emphasises flexibility and collaboration	P1
Waterfall Model	Sequential approach to software development with distinct phases	P1
User Acceptance Testing	Final testing phase where end users verify the system meets their needs	P1
Feasibility Study	Assessment of the practicality and viability of a proposed project	P1
Data Protection Act	Legislation governing the collection, use, and storage of personal data	P1
GDPR	General Data Protection Regulation — EU law on data protection and privacy	P1
Computer Misuse Act	UK legislation criminalising unauthorised access to computer systems	P1

Term	Definition	Paper
Health and Safety	Regulations protecting workers from hazards including RSI, eye strain, and posture issues	P1
Green Computing	Environmentally responsible design, manufacture, use, and disposal of computers	P1
Teleworking	Working from a location other than the traditional office using ICT	P1
Video Conferencing	Live visual communication between people in different locations	P1
Expert System	Computer system that emulates the decision-making ability of a human expert	P1
Robotics	Design, construction, and operation of robots using computer systems	P1
Artificial Intelligence	Simulation of human intelligence processes by computer systems	P1
Machine Learning	Subset of AI where systems learn from data without being explicitly programmed	P1
Neural Network	Computing system inspired by biological neural networks, used in deep learning	P1
Big Data	Extremely large datasets that require specialised tools and methods to analyse	P1
Data Mining	Process of discovering patterns in large datasets using machine learning and statistics	P1

How to use this glossary: Students should review 10 terms per week, creating flashcards or using digital tools (Quizlet, Anki). Terms marked P1 are theory-based and tested in Paper 1; P2 and P3 terms are practical skills tested in Papers 2 and 3.

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